

Physics-Informed Identification Limits of Thermal Dynamics Under Degraded Telemetry in High-Performance Computing Systems

Complete Implementation Guide and Publication-Safe Research Blueprint

One-sentence paper idea: Use real Summit supercomputer power/temperature telemetry, deliberately degrade the temperature measurements in controlled ways, and compare classical identification, data-driven neural models, and a physics-informed neural network (PINN) to determine whether physics constraints preserve reliable thermal-parameter identification when telemetry quality falls.

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Scope: implementation, novelty safety, experiments, equations, evaluation, risks, and publication strategy

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This document is a research blueprint, not a claim of guaranteed publication. Novelty statements should be rechecked immediately before submission.

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1 Executive Summary

Recommended paper: *Physics-Informed Identification Limits of Thermal Dynamics Under Degraded Telemetry in High-Performance Computing Systems.*

The paper should not be framed as “another PINN that predicts temperature.” That contribution is now crowded. The stronger question is:

How far can thermal measurements be reduced or degraded before the physical thermal dynamics of an HPC system can no longer be identified reliably, and can PINN-based learning extend that reliable regime?

The core output is therefore not only a temperature error score. It is a **thermal identifiability map** linking telemetry quality to the reliability of recovered parameters such as the thermal time constant τ and power-to-temperature gain K .

1.1 Recommended contribution hierarchy

1. **Primary:** quantify thermal parameter-identification reliability under controlled telemetry degradation on real HPC data.
2. **PINN contribution:** test whether physics constraints improve parameter recovery and physical consistency relative to OLS, MLP, and LSTM baselines.
3. **Boundary contribution:** estimate the transition from reliable to unreliable identification as sensor coverage, sampling, and quantization worsen.
4. **Application contribution:** derive practical telemetry-design guidance for monitoring, storage, communication, and thermal digital twins.

2 What Exactly Is the Paper?

2.1 Problem

Modern HPC systems expose large streams of power and temperature telemetry. Summit’s public dataset contains long-term per-component power and thermal measurements derived from 1 Hz telemetry, with released 10-second and 1-minute aggregate datasets. The public repository reports 4,626 compute hosts, CPU/GPU power and temperature variables, and large sample subsets suitable for analysis. [1, 2]

The research problem is that a thermal model identified from telemetry can be affected by:

- missing or sparse sensor observations;
- temporal subsampling;
- limited sensor resolution such as quantization;
- changing workload and operating conditions;
- model mismatch between a simple lumped thermal law and the real machine.

2.2 Physical model

Start with a deliberately simple, interpretable lumped thermal dynamic model:

$$\frac{dT}{dt} = -\frac{1}{\tau}(T - T_a) + KP, \quad (1)$$

where T is component temperature, T_a is an ambient or reference cooling temperature, P is power input, $\tau > 0$ is a thermal time constant, and $K > 0$ converts power to temperature rise.

This model is intentionally not presented as a complete physical description of Summit. It is the **identification target**. The real system contains nonlinearities, multi-node heat paths, direct liquid cooling, workload-dependent effects, and unmeasured states.

2.3 PINN formulation

A continuous-time PINN approximates the temperature trajectory $\hat{T}_\theta(t, P, T_a)$ while the physics parameters are learned jointly:

$$r_\theta(t) = \frac{d\hat{T}_\theta}{dt} + \frac{1}{\tau}(\hat{T}_\theta - T_a) - KP. \quad (2)$$

The physics residual loss is

$$\mathcal{L}_{phys} = \frac{1}{N_f} \sum_{j=1}^{N_f} r_\theta(t_j)^2. \quad (3)$$

For observed data $\{t_i, T_i^{obs}, P_i, T_{a,i}\}$,

$$\mathcal{L}_{data} = \frac{1}{N_o} \sum_{i=1}^{N_o} (\hat{T}_\theta(t_i) - T_i^{obs})^2. \quad (4)$$

The basic total loss is

$$\mathcal{L} = \lambda_d \mathcal{L}_{data} + \lambda_p \mathcal{L}_{phys} + \lambda_{ic} \mathcal{L}_{IC} + \lambda_r \mathcal{L}_{reg}. \quad (5)$$

Critical modeling warning: Do not build a neural residual that is so unconstrained that it can absorb any modeling error while the learned τ and K become arbitrary. That can produce excellent temperature prediction and physically meaningless parameters. The physics-based ODE must remain explicit and the learned correction, if used, must be separately controlled and reported.

3 What Has Already Been Done?

3.1 Thermal PINNs are already established

PINNs are now common in heat transfer and inverse thermal problems. Recent work includes sparse thermal reconstruction, inverse heat conduction, thermal-property estimation, and data-center thermal modeling. This means “PINN + heat transfer” is not enough novelty.

3.2 Sparse-sensor thermal identification already exists

A 2026 International Journal of Heat and Mass Transfer paper uses a hybrid PINN for a 2D transient inverse heat-conduction problem and explicitly studies limited temperature measurements, orthotropic-property identification, heat-source characterization, and physics-guided sensor selection using observability and parameter sensitivity. [4]

A 2026 MATLAB Central implementation also describes sparse-sensor PINN reconstruction and joint identification of thermal diffusivity, heat loss, and source amplitude from noisy point measurements on a 2D benchmark. [6]

A 2025 Engineering Applications of Artificial Intelligence study reported PINN-based thermal property identification from limited measurements, reinforcing that sparse-data inverse thermal identification is an active topic. [7]

3.3 Quantized thermal identification is no longer a safe standalone novelty

A 2026 Applied Thermal Engineering paper titled “Motor temperature signal reconstruction and wide-temperature-range identification based on a physics-constrained neural network” uses coarse, quantized stepwise temperature measurements and a mechanism-data hybrid network to reconstruct trajectories and identify thermal parameters. It explicitly processes 1 C quantized automotive sensor data and uses an autoregressive residual network with an embedded energy-balance equation. [5]

Therefore, a paper whose entire contribution is simply “PINN under 1 C / 2 C / 5 C quantization” would be too close to existing work.

3.4 Data-center thermal PINNs already exist

A 2025 International Communications in Heat and Mass Transfer article develops hybrid PINNs for airflow and thermal modeling in data-center environments, including sparse labeled data and parameterized operating conditions. [3]

Therefore, do not claim that the paper is the first PINN for data-center thermal modeling.

3.5 Summit telemetry analysis already exists

The Summit dataset itself is accompanied by an SC’21 paper analyzing power, energy, thermal behavior, jobs, and GPU failures at very large scale. [2]

That work is not a PINN thermal-identification study. This creates the opportunity to turn a well-characterized production telemetry source into an inverse, physics-informed learning benchmark.

4 Where Is the Defensible Novelty?

Use this novelty framing: “To our knowledge, existing studies have separately addressed sparse thermal inverse identification, quantized thermal signals, data-center PINNs, and large-scale HPC telemetry. We investigate their intersection by quantifying thermal-dynamics identifiability under controlled degradation of production HPC telemetry and by testing whether physics-informed learning changes the practical identifiability boundary.”

Avoid “first ever,” “solves the identifiability problem,” and “recovers the true thermal constant” unless the experimental design can actually support those claims.

5 What Needs To Be Done?

5.1 Main research question

RQ1: How do sensor sparsity, temporal degradation, and temperature quantization affect the identification of thermal dynamics in HPC systems?

RQ2: Does a PINN recover interpretable thermal parameters more reliably than non-physics baselines under the same degraded observations?

RQ3: Can we identify a practical boundary separating reliable from unreliable parameter identification?

5.2 Main hypothesis

PINN physics constraints should reduce parameter bias and variance under moderate telemetry degradation, but physics cannot recover information that the measurements no longer contain. Therefore, PINN performance should improve the reliability frontier rather than eliminate the identifiability limit.

6 Safe Implementation Strategy

6.1 Stage 0 - Synthetic benchmark first

Before touching Summit, generate synthetic trajectories from Eq. (1) with known τ and K . Add the exact degradation mechanisms you intend to study. This gives known ground truth and answers the reviewer question: “How do you know parameter recovery is correct?”

Suggested synthetic setup:

- choose several ground-truth pairs, e.g. $\tau \in \{10, 30, 60, 120\}$ s and K values scaled to realistic temperature rises;
- drive the system with step, ramp, stochastic, and workload-like power inputs;
- generate a high-resolution reference trajectory;
- create degraded observations from that reference.

6.2 Stage 1 - Summit reference dataset

Use the public Summit 10-second sample initially. The repository reports a 58-host full-period sample of about 4.6 GB and a 100-host sample of about 4.2 GB. The full 10-second dataset is approximately 492 GB and the 1-minute dataset about 120 GB. [1]

Start with 58 hosts because it is tractable on a student workstation or modest workstation/server setup. Scale later if the results are stable.

6.3 Stage 2 - Define a node/component thermal channel

The cleanest first implementation is one component at a time, for example:

$$T_t = \text{gpu0_core_temp}_t, \quad P_t = \text{p0_gpu0_power}_t. \quad (6)$$

Alternative channels can be used for robustness:

- GPU core temperature vs GPU power;
- CPU mean core temperature vs CPU package power;
- hottest component temperature as a stress case.

Do not mix CPU and GPU dynamics in the first experiment. Their thermal structures are not guaranteed to share one τ .

6.4 Stage 3 - Data preprocessing

1. Parse timestamps and sort by host and time.
2. Remove powered-off or invalid periods.
3. Drop or segment around long missing intervals.

4. Create a fixed sampling interval within each training segment.
5. Normalize t , P , and T for stable neural optimization.
6. Preserve an unmodified reference copy of the data before any degradation.

The Summit repository documents missing-value handling and reports that released data were selected from periods with minimal data loss relative to operational interruptions. [1]

7 Telemetry Degradation Protocol

7.1 A. Sensor sparsity

Mask observed temperature samples at controlled coverage levels:

$$100\%, 75\%, 50\%, 25\%, 10\%, 5\%.$$

Use at least three masking strategies:

1. uniform random masking;
2. contiguous blocks, simulating outages;
3. periodic masking, simulating reduced reporting frequency.

The test target should remain the original reference temperature for evaluation, but the model should only see the masked observations.

7.2 B. Temporal degradation

Because the public Summit release is based primarily on 10-second and 1-minute aggregate measurements rather than the original 1 Hz stream, do not pretend that the public data allow direct validation of sub-second effects. The dataset documentation explicitly states that the public primary sets are 10-second and 1-minute means derived from original 1 Hz measurements. [1]

You can still create controlled temporal experiments within the public 10-second data, for example:

$$10s \rightarrow 20s \rightarrow 40s \rightarrow 60s.$$

For each level, aggregate or subsample consistently.

7.3 C. Quantization

Apply synthetic temperature quantization after preserving the full-precision reference:

$$q \in \{0, 0.5, 1, 2, 5\} \text{ } ^\circ\text{C}.$$

For a scalar temperature T :

$$T_q = q \cdot \text{round}(T/q). \tag{7}$$

Use quantization as one degradation axis, not as the entire novelty claim.

7.4 D. Combined degradation

For the strongest robustness experiment combine:

- 25% or 10% sensor coverage,
- 20-60 s effective sampling,

- 1-2 C quantization.

Do not explode the number of combinations before the core result is stable. Humans have a peculiar habit of generating 200 plots and then discovering that only three actually answer the research question.

8 Model Implementations

8.1 Baseline 1 - OLS / discrete thermal identification

From

$$T_{t+1} = \alpha T_t + \beta P_t + \gamma,$$

fit α, β, γ by least squares. Convert α to an approximate time constant:

$$\tau = -\frac{\Delta t}{\ln(\alpha)}, \quad 0 < \alpha < 1. \quad (8)$$

Report this as a reference mechanistic baseline, not absolute truth.

8.2 Baseline 2 - MLP predictor

Train a standard feed-forward model to predict T_{t+1} from lagged temperatures and power. This tests whether the benefit is simply nonlinear regression.

8.3 Baseline 3 - LSTM

Use a short sequence of past temperatures and power values. Keep architecture and tuning budget comparable with the PINN.

8.4 Primary model - continuous-time inverse PINN

Recommended network input:

$$(t, P, T_a, \text{optional operating covariates})$$

and output:

$$\hat{T}_\theta(t).$$

Make τ and K trainable positive parameters:

$$\tau = \exp(\eta_\tau), \quad K = \exp(\eta_K). \quad (9)$$

Autodifferentiation supplies $d\hat{T}/dt$.

8.5 Optional stronger hybrid model

After the strict inverse PINN works, introduce a bounded discrepancy term:

$$\frac{dT}{dt} = -\frac{1}{\tau}(T - T_a) + KP + g_\phi(T, P, h), \quad (10)$$

where g_ϕ is a small neural correction.

Constrain g_ϕ with a strong regularizer, bounded output, or low-complexity architecture. Report the strict PINN and hybrid model separately. The strict PINN is safer for physical interpretability.

9 Recommended PINN Training Procedure

1. Select one host/component trajectory and split by time into train, validation, and held-out test windows.
2. Initialize τ from the OLS baseline, while also running multiple random initializations.
3. Train the network and τ, K jointly.
4. Monitor $\mathcal{L}_{data}$ and $\mathcal{L}_{phys}$ separately.
5. Use early stopping on the validation objective.
6. Repeat across at least 5 random seeds for each important degradation point.
7. Bootstrap over time windows and hosts to estimate variability.

9.1 Avoid leakage

A time-series split must not randomly shuffle consecutive points across train and test. Prefer:

earlier time $\rightarrow$ train, middle $\rightarrow$ validation, later $\rightarrow$ test.

For cross-host generalization, train on one set of hosts and hold out different hosts.

10 Implementation Pseudocode

```
# X = normalized [t, power, ambient]
# y = observed temperature at available sensor points

net = TemperaturePINN()
eta_tau = Parameter(log(tau_init))
eta_K = Parameter(log(K_init))
optimizer = Adam(list(net.parameters()) + [eta_tau, eta_K], lr=1e-3)

for epoch in range(num_epochs):
    T_hat = net(t_obs, power_obs, ambient_obs)
    loss_data = mse(T_hat, T_obs)

    t_f = collocation_times.requires_grad_(True)
    T_f = net(t_f, power_f, ambient_f)
    dTdt = grad(T_f, t_f, create_graph=True)

    tau = exp(eta_tau)
    K = exp(eta_K)
    residual = dTdt + (T_f - ambient_f) / tau - K * power_f
    loss_phys = mean(residual**2)

    loss_ic = (net(t0, p0, a0) - T0)**2
    loss_reg = small_prior_penalty(tau, K)

    loss = (lam_data * loss_data
            + lam_phys * loss_phys
            + lam_ic * loss_ic
            + lam_reg * loss_reg)

    optimizer.zero_grad()
    loss.backward()
    clip_grad_norm_(all_trainable_parameters, max_norm=1.0)
    optimizer.step()

\n# Evaluate only after training:
# temperature metrics, tau/K error, residuals, uncertainty.
```

11 What Should Count as “Identification”?

Prediction and identification are separate objectives.

11.1 Prediction metrics

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (\hat{T}_i - T_i)^2}, \quad (11)$$

$$R^2 = 1 - \frac{\sum_i (\hat{T}_i - T_i)^2}{\sum_i (T_i - \bar{T})^2}. \quad (12)$$

11.2 Parameter recovery

If ground truth is available in synthetic data:

$$E_\tau = \frac{|\hat{\tau} - \tau^*|}{\tau^*} \times 100\%. \quad (13)$$

For Summit, do not call the full-data estimate “true.” Use terms such as “reference estimate” or “high-quality-data estimate.”

11.3 Uncertainty and stability

Report:

- median relative τ error;
- interquartile range;
- bootstrap 95% confidence interval;
- coefficient of variation across seeds/windows;
- parameter bias where synthetic ground truth exists.

12 Defining the Identifiability Boundary Safely

Do not declare a boundary from one arbitrary score.

A practical definition can combine two criteria:

1. relative parameter error below a predeclared tolerance, e.g. 10%;
2. confidence interval width below a predeclared fraction of the reference estimate.

Run sensitivity analysis over several thresholds such as 5%, 10%, and 20%. If the qualitative ranking is stable, the conclusion is much stronger.

The boundary is then the largest degradation level for which the model satisfies the chosen reliability criteria.

Important: “PINN still predicts temperature accurately” does not imply “PINN identified the true thermal parameter.” The entire paper should explicitly separate predictive accuracy from physical parameter identifiability.

13 Core Experimental Matrix

Experiment	Purpose	Conditions	Main output
Synthetic known-parameter	Verify inverse PINN	several τ, K , several power profiles	true parameter error
Full Summit reference	establish real-world baseline	clean 10 s data	reference τ, K and prediction
Sensor sparsity	test missing observations	100,75,50,25,10,5%	error vs coverage
Temporal degradation	test reduced reporting rate	10,20,40,60 s	error vs interval
Quantization	test sensor resolution	0,0.5,1,2,5 C	error vs resolution
Combined degradation	stress test	selected combinations	reliability map
Cross-host test	test generalization	unseen hosts	held-out-host metrics
Ablation	prove PINN mechanism	no physics / no parameter learning / no IC	performance drop

14 Ablation Studies That Reviewers Will Expect

1. Data loss only vs data + physics loss.
2. Fixed τ, K vs trainable τ, K .
3. Strict PINN vs hybrid residual PINN.
4. No quantization vs quantized measurements.
5. Random vs block sensor dropout.
6. One host vs multiple hosts.
7. Single seed vs multiple seeds.

15 Figures That Should Drive the Paper

1. **Conceptual diagram:** Summit telemetry -> controlled degradation -> OLS/MLP/LSTM/PINN -> temperature + parameter recovery -> identifiability map.
2. **Representative trajectory:** reference vs degraded observations vs PINN reconstruction.
3. **Parameter recovery plot:** relative τ error vs sensor coverage for every model.
4. **Heatmap:** sensor coverage vs quantization level, colored by parameter reliability.
5. **Residual plot:** physical residual distribution for PINN vs data-only network.
6. **Uncertainty plot:** bootstrap CI width vs degradation.
7. **Cross-host plot:** distribution of τ errors across held-out hosts.
8. **Failure case:** an example where all models fail, demonstrating that physics does not magically create missing information.

16 Practical Application of the Final Result

The paper becomes application-oriented when it translates model behavior into telemetry guidance. For example, the final study could report a rule of the form:

Under the tested workload and hardware subset, temperature parameter identification remained within the predefined reliability region down to approximately X% effective observations and Y C quantization, while below that region the confidence interval and parameter bias grew rapidly.

The exact X and Y must come from the experiment. Never manufacture them in the abstract before running the study.

Potential uses include telemetry design, monitoring bandwidth reduction, storage planning, thermal digital twins, anomaly detection, and deciding when reduced sensing is safe enough for model-based control.

17 What We Can Add for a Stronger Paper

17.1 Extension 1 - Multi-node thermal model

Replace the scalar ODE with a low-order thermal network:

$$\mathbf{C}\dot{\mathbf{T}} = \mathbf{A}\mathbf{T} + \mathbf{B}P + \mathbf{q}. \quad (14)$$

Learn physically constrained conductances/capacitances. This increases realism but also increases identifiability difficulty.

17.2 Extension 2 - Workload regime conditioning

Add a latent or observed operating-regime variable h to the model:

$$\dot{T} = f_{phys}(T, P, T_a; \tau, K) + g_{\phi}(T, P, h). \quad (15)$$

This can capture workload-dependent behavior without abandoning physical interpretability.

17.3 Extension 3 - Physics-guided sensor selection

Once the basic paper is stable, use sensitivity or Fisher-information-style diagnostics to identify which sensors contribute most to parameter recovery. This links directly to recent literature on physics-guided observability and sensitivity. [4]

17.4 Extension 4 - Uncertainty-aware PINN

Use ensembles or Bayesian approximations to report credible uncertainty rather than a single deterministic estimate.

18 What Not To Do

1. Do not use only RMSE and claim thermal identification.
2. Do not call OLS from full Summit data “ground truth.”
3. Do not claim the first sparse-sensor thermal PINN.
4. Do not make quantization the entire novelty.
5. Do not use random train/test splitting across time.
6. Do not let a high-capacity residual network absorb the entire governing equation.
7. Do not begin with the entire 492 GB Summit dataset. Start with the 58-host sample.
8. Do not claim the PINN always wins. A scientifically credible paper needs failure cases.

19 Recommended Software Stack

- Python 3.x
- PyTorch
- NumPy / SciPy
- Pandas for small subsets; Dask and PyArrow for larger Summit parquet data, consistent with the dataset documentation. [\[1\]](#)
- Matplotlib for figures
- scikit-learn for OLS and standard metrics
- Optional: DeepXDE or NVIDIA Modulus as a reference implementation, but PyTorch gives maximum control over inverse parameters and losses.

20 Compute Plan

20.1 MVP

A CPU workstation can run the synthetic and small Summit experiments. Begin with one component per host and short windows.

20.2 Recommended

A single modern NVIDIA GPU is sufficient for initial PINN training. Multiple independent runs can be parallelized across CPU processes or GPUs.

20.3 Scale-up

Use Dask/Parquet filtering to avoid materializing the complete 492 GB dataset. Run host-level experiments independently and aggregate statistics afterward.

21 Reproducibility Protocol

Store:

- raw data identifier and dataset DOI;
- exact host list;
- preprocessing configuration;
- degradation seed and masking rule;
- model configuration;
- loss weights;
- random seeds;
- trained parameter estimates;
- metrics and confidence intervals;
- environment file and code commit.

Every final plot should be reproducible from a saved results table.

22 Minimum Viable Publishable Study

If time or compute becomes limited, do this exact subset:

1. Synthetic known-parameter validation.
2. Summit 58-host sample.
3. One GPU temperature + corresponding GPU power channel.
4. OLS, MLP, LSTM, strict inverse PINN.
5. Sensor coverage: 100, 50, 25, 10, 5%.
6. Quantization: 0, 1, 2 C.
7. 5 random seeds at the key points.
8. Temperature prediction + τ error + bootstrap uncertainty.
9. One final reliability heatmap.

This is already a coherent paper if the results are strong and the claims are disciplined.

23 Strong Version

For a more competitive journal submission, add:

1. block/periodic sensor failures;
2. temporal degradation;
3. cross-host generalization;
4. two hardware channels, e.g. CPU and GPU;
5. a low-order multi-node thermal model;
6. sensitivity-based observability analysis;
7. uncertainty-aware PINN or ensemble PINN;
8. a practical telemetry recommendation derived from the identifiability frontier.

24 Proposed Paper Structure

1. Introduction and motivation
2. Related work: PINNs, inverse thermal identification, sparse sensing, HPC telemetry
3. Problem formulation
4. Summit dataset and preprocessing
5. Physics-informed thermal identification method
6. Controlled telemetry degradation framework
7. Experimental design and baselines
8. Results: prediction
9. Results: parameter identification
10. Results: identifiability boundary and uncertainty
11. Discussion: what physics can and cannot recover
12. Practical telemetry implications
13. Limitations and future work
14. Conclusion

25 Draft Abstract Blueprint

Background: Thermal models of high-performance computing systems are increasingly identified from telemetry generated by the machines themselves, yet practical telemetry is finite, aggregated, and potentially degraded.

Objective: This work investigates how sensor sparsity and measurement degradation affect the identification of thermal dynamics in a production HPC system and whether physics-informed learning can preserve physically meaningful parameter estimates under reduced observations.

Method: We develop an inverse PINN based on a first-order thermal dynamic model and jointly estimate temperature trajectories and interpretable thermal parameters. Using the publicly available Summit supercomputer power and temperature telemetry, we construct controlled degradation conditions spanning sensor sparsity, temporal reduction, and temperature quantization. OLS, MLP, and LSTM baselines are evaluated under identical observation conditions. Synthetic trajectories with known parameters are additionally used to quantify absolute identification error and uncertainty.

Results to target, not invent: The study will quantify temperature prediction error, thermal time-constant bias, variability across hosts and seeds, and the degradation point at which parameter identification becomes unreliable.

Contribution: Rather than proposing PINNs solely as another temperature predictor, the paper establishes a practical framework for measuring the telemetry quality required for physically reliable thermal identification in HPC systems and tests whether physics constraints extend that reliable region.

Application: The resulting identifiability map can guide telemetry reduction, sensor configuration, storage planning, and thermal digital-twin development.

26 Publication-Safe Claims

Recommended claims:

- “We quantify thermal parameter-identification robustness under controlled degradation of real HPC telemetry.”
- “We compare physics-informed and data-only approaches under matched observation budgets.”
- “We identify a practical reliability frontier rather than assuming physics can recover unobservable information.”
- “We demonstrate the method on publicly available Summit production telemetry.”

Avoid:

- “PINNs solve the thermal identifiability problem.”
- “This is the first study of sparse thermal PINNs.”
- “Our Summit parameters are the true physical constants.”
- “PINNs always outperform LSTM/OLS.”

27 Risk Register and Mitigation

Risk	Why it matters	Mitigation
Model mismatch	First-order ODE may be too simple	present it as low-order identification target; add controlled hybrid correction as extension
No true τ on Summit	absolute parameter error is unknowable	synthetic ground truth + Summit reference estimates
PINN instability	training can be sensitive to scaling/loss weights	normalize inputs, initialize from OLS, multiple seeds, report failures
Data leakage	time-series leakage inflates generalization	chronological splits and held-out hosts
Sparse sensors already studied	novelty becomes weak	focus on production HPC telemetry and identifiability frontier
Quantization already studied	overlap with recent PCNN work	treat quantization as one axis within broader degradation framework
Too much computation	full dataset is huge	start with 58-host sample and filter with Parquet/Dask
Overclaiming physics benefit	reviewer may reject weak causality	ablations separating prediction gain from parameter-identification gain

28 Final Go / No-Go Checklist

28.1 GO if

- synthetic known-parameter tests recover τ and K consistently;
- PINN is physically constrained and interpretable;
- Summit experiments show a systematic degradation trend;

- at least one result demonstrates a meaningful difference between prediction accuracy and parameter identifiability;
- the reliability boundary is stable across seeds/hosts.

28.2 NO-GO / redesign if

- PINN only improves RMSE but not parameter reliability;
- learned parameters vary wildly under small perturbations;
- the physical ODE residual can be reduced only by making the network effectively unconstrained;
- results are indistinguishable from OLS and the paper has no additional insight;
- all degradation levels remain trivial, producing no meaningful boundary.

29 Final Recommendation

Commit to one paper: *Physics-Informed Identification Limits of Thermal Dynamics Under Degraded Telemetry in High-Performance Computing Systems.*

Treat the study as an **inverse PINN paper with a telemetry-reliability question**, not as a generic temperature-prediction paper.

The safest scientific sequence is:

synthetic ground truth → Summit reference → sensor degradation → PINN + baselines → parameter recovery →

That sequence gives the project three things reviewers actually care about: a real problem, a clearly motivated use of PINNs, and a measurable contribution beyond “our neural network had a smaller RMSE.”

30 Sources and Literature Notes

References

- [1] Shin, W. et al. *Long Term Per-Component Power and Thermal Measurements of the OLCF Summit System*. Oak Ridge National Laboratory dataset, DOI 10.13139/OLCF/1861393. Public dataset documentation and companion repository. <https://doi.org/10.13139/OLCF/1861393>
- [2] Shin, W., Oles, V., Karimi, A. M., Ellis, J. A., Wang, F. *Revealing Power, Energy and Thermal Dynamics of a 200PF Pre-Exascale Supercomputer*. SC '21, ACM, DOI 10.1145/3458817.3476188.
- [3] Hashemi, Z., Gholampour, M., Su, H.-H., Chiang, C.-Y., Chang, H.-H., Wang, C.-C. *Hybrid physics-informed neural networks for airflow and thermal modeling in data center environments*. International Communications in Heat and Mass Transfer, 168, 109511, 2025. DOI 10.1016/j.icheatmasstransfer.2025.109511.
- [4] *Physics-informed neural networks for 2D-transient inverse heat conduction problem involving static and moving heat source*. International Journal of Heat and Mass Transfer, 259, 128349, 2026. DOI 10.1016/j.ijheatmasstransfer.2026.128349.
- [5] Xu, N., Zhang, Z., Zhang, Q., Ji, H., Li, T. *Motor temperature signal reconstruction and wide-temperature-range identification based on a physics-constrained neural network*. Applied Thermal Engineering, DOI 10.1016/j.applthermaleng.2026.130703.

- [6] César. *Sparse-Sensor PINN Reconstruction and Identification*. MATLAB Central File Exchange, version 3.0.0, July 2026. <https://www.mathworks.com/matlabcentral/fileexchange/183734-sparse-sensor-pinn-reconstruction-and-identification>
- [7] *Physics-informed neural networks for thermal-property identification with limited measurements*. Engineering Applications of Artificial Intelligence, 2024. Search record used for literature-positioning; exact bibliographic metadata should be verified before journal submission.